

BTP610 Test 3 Reference Sheet

Week 8 React Native Maps | Week 9 Expo Location | Week 10 Firebase MCQ

Week 8 Setup

- Install: `npm install react-native-maps`
- Import: `MapView, Marker, Region`
- React: `useRef, useState`
- Ref syntax: `useRef<MapView>(null)`

Week 8 Region

- `Region = { latitude, longitude, latitudeDelta, longitudeDelta }`
- `latitude, longitude`: center point
- `latitudeDelta, longitudeDelta`: zoom area
- Smaller delta = more zoom

Week 8 Map Syntax

- `<MapView initialRegion=... />`
- `<MapView region=... />`
- `initialRegion`: starting map only
- `region`: controlled map from state
- `ref={mapRef}`
- Optional prop: `zoomControlEnabled={true}`

Week 8 Marker Syntax

- `<Marker coordinate={{ latitude, longitude }} />`
- Common props: `title, description`
- Array render pattern: `markerList.map(...)`
- Coordinate keys must be `latitude/longitude`

Week 8 Actions

- `mapRef.current?.animateToRegion(region)`
- Add marker: update state array with spread
- Clear markers: set state to empty array
- Buttons use `TouchableOpacity + onPress`

Week 8 Likely Tasks

- show map with starting region
- add random marker
- clear all markers
- zoom to selected location
- render markers from state

Week 9 Setup

- Install: `npx expo install expo-location`
- Also need maps package
- Import namespace: `* as Location`
- Often paired with `MapView, Marker, Region`

Week 9 State Shapes

- `Region | null`
- `address: string`
- `currentAddress: string`
- Current coords come from `location.coords`

Week 9 Permission Flow

- `Location.requestForegroundPermissionsAsync()`
- Read status
- If `status !== "granted"`: stop and alert
- Permission required before current location

Week 9 GPS Syntax

- `Location.getCurrentPositionAsync(...)`
- Accuracy option: `Location.Accuracy.High`
- Build `Region` from `location.coords.latitude` and `location.coords.longitude`
- Set region state after reading coords

Week 9 Geocoding

- `Location.geocodeAsync(address)`
- Meaning: `address → coordinates`
- Read: `result[0].latitude, result[0].longitude`
- Check `result.length` before using values

Week 9 Reverse Geocoding

- `Location.reverseGeocodeAsync({ latitude, longitude })`
- Meaning: `coordinates → readable address`
- Common field: `formattedAddress`
- Useful after getting current location

Week 9 Map Syntax

- Conditional render: map only when `region` exists
- Marker `coordinate` often uses `region.latitude` and `region.longitude`
- Marker `description` can show current address
- Input uses `TextInput, value, onChangeText`

Week 9 Likely Tasks

- ask for location permission
- get current location
- show current location on map

- search typed address
- show readable address after reverse geocode

Week 9 Common Errors

- forgetting permission request
- mixing geocode and reverse geocode
- not checking `result.length`
- rendering map before `region` exists
- forgetting deltas in a `Region`

Week 10 Core Terms

- Install: `npm install firebase`
- `initializeApp`: start Firebase app
- `getAuth`: authentication service
- `getFirestore`: Firestore database service

Week 10 Imports

- from `"firebase/app"`: `initializeApp`
- from `"firebase/auth"`: `getAuth, onAuthStateChanged`
- from `"firebase/firestore"`: `getFirestore`

Week 10 Auth Terms

- `createUserWithEmailAndPassword`
- `signInWithEmailAndPassword`
- `signOut`
- `User | null`
- `onAuthStateChanged`

Week 10 Route Guard

- If no user: redirect to sign in
- If user exists: redirect to app screen
- Expo Router term used in sample: `Redirect`

Week 10 Firestore Terms

- Firestore = cloud NoSQL database
- Collection name example: `"books"`
- Common CRUD terms: `addDoc, getDocs, updateDoc, deleteDoc`
- For your test, Firebase is MCQ only

High-Yield MCQ Facts

- `MapView` = map component
- `Marker` = pin component
- `Region` needs 4 values
- `geocodeAsync`: `address → coords`
- `reverseGeocodeAsync`: `coords → address`
- `onAuthStateChanged`: listens for auth changes